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Generalized Stochastic Workflow Net-based Quantitative Analysis of Business Process Performance

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***Abstract* - During the life cycle of business process management, the quantitative analysis of the time performance measure is pivotal to guarantee the validity of process model and to improve its quality. Based on the characteristic of management processes and the assumption that the processing time of activities is either zero or a stochastic variable obeying the exponential distribution, the Generalized Stochastic Workflow Net model of the business process is constructed, which facilitates the analysis of time performance through the theories on Generalized Stochastic Petri Net and Markov Chain. Then the computational formula of the average flow time of workflow is deduced. Finally, a case study is given, and the computational value is compared with that of simulation. The test shows that these two results are almost identical.**

Index Terms - Business process management; Generalized Stochastic Workflow Net; Performance analysis

I. INTRODUCTION

In recently, BPM (Business process management) has been a hotspot in the research fields of EAI (Enterprise Application Integration) and ERP (Enterprise Resources Planning). Originating from workflow management, BPM aims at supporting and managing business processes entirely by using lots of new methods, techniques and workflow software to design, enact, control and analyze practical operational processes involving humans, organizations, enterprise applications, business objects and other information resources [1]. Unlike the studies in workflow management which mainly focuses on the automation of business processes and their software implementation, in BPM the analysis and optimization of the workflow performance is much more emphasized. It is reported that the sojourning time of the non-value-add activities accounts for up to 90% of the throughout time, which means that currently the waste of resources and time in business processes is very large. As a result, if 50% of the non-value-add time can be reduced by business process reengineering, its significance is much higher than the reduction of the value-add time to 50% by the technical innovation.

The performance analysis, often called BPA (Business Process Analysis), is critical to process optimizing and reengineering. In general, there are three types of dynamic running process information need to be collected and analyzed, which are the time constraints, the utilization efficiency of the resources and the time performance [2]. At present, most research interests focused on the former two areas. For example, Eder et al. [3] classified the time sequence constraints into different types and utilized the time constrained Petri net to describe and validate these constraints. Li et al. [4] put forward a transition component net and applied it into the validation of the time boundedness. Hee et al. [5] discussed the resource allocation strategy and proposed an optimized allocation algorithm for the complex workflow. However, there is still no powerful and convenient quantitative method to compute the time performance indices of business process, which are often obtained by simulation yet [2]. Since the modelling and maintenance of the simulation model is always complex, it is not practical for the common users to set up a simulation model and obtain the statistic data by lots of simulation when the model are changed.

In nature, the time performance computation of business process belongs to one problem of performance analysis for the DEDS (Discrete Event Dynamic System). Chen et al. [6] applied the Stochastic Petri Net model to analyze the time performance. Zerguini et al. [7] combined Stochastic Petri Net and Laplace Transformation to compute the throughput time of workflow. These researches enlighten our work.

In this paper, we apply the theory of Generalized Stochastic Petri Net and Markov Chain into the performance analysis of business process and set up a formula to calculate the mean throughout time. The remainder is organized as follows. First, the basic theory about the Generalized Stochastic Workflow Net is introduced. Then the calculation formula of mean throughout time is put forward. Finally, a case study from a telecommunication company is given to illustrate the whole procedures. The result is also compared with that of simulation.

II. GENERALIZED STOCHASTIC WORKFLOW NET

Workflow is defined as the totally or partly automatic execution of the enterprise business processes [8]. It takes a few intervals for the activities within a workflow model to be processed. If its processing time is zero, the activity is called the immediate task otherwise is called the time-delay task. In the Petri-net-based model [9], the activities are depicted as

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transitions. When the temporal factor is taken into account, a time-delay variable should be attached onto each transition, thus the so-called Timed Petri Net is emerged. Due to the uncertainty in the processing time, such a time-delay variable is often not determinately but stochastic, which makes the Petri net model be a stochastic one.

Definition 1 (SPN). The Stochastic Petri Net [10] is denoted as a 2-tuple $<PN, F>$, where PN is a classical Petri net, and F is the stochastic time-delay function of the marking M and the transition t. That is, the variable $F(M, t)$ is the interval that the transition t in the marking state of M spend in firing.

Due to the diversity in the type of workflow and task, $F(M, t)$ may obey different stochastic distributions. For most of the management workflows, the arrival of workflow or task instances is often considered as a Poisson flow, of which the interval between two instances has the memoryless character [8]. Therefore, we assume that the processing time of each activity obeys the exponential distribution $exp(\lambda)$.

Definition 2 (GSPN). The Generalized Stochastic Petri Net [10] is one special class of Stochastic Petri Nets such that $GSPN=(PN,\Lambda,S)$, where PN is a classical Petri net. The transition set T is divided into two child sets, one is the immediate transition set T_I and the other is the exponential transition set T_E, such that, $T=T_I+T_E$, $T_I=\{T_{I1},\ldots,T_{Im}\}$, $T_E=\{T_{E1},\ldots,T_{En}\}$. The mean firing speeds of all exponential transitions form a set: $\Lambda=\{\lambda_{E1},\ldots,\lambda_{En}\}$. S is a set of random switches, which is composed of all enabled and conflicting transitions endued with some probability distribution. S can be empty if no branch structure exists in the process model.

Every GSPN model consists of several different markings M. It is obvious that the transformation process of markings is a stochastic process. There are two types of marking. One is the tangible marking, which indicates that in this state the net system will stay for a non-zero period of time before one of the transitions is fired. The other is the vanishing marking, which indicates that the firing process is finished immediately.

Combining the Generalized Stochastic Petri Net with the Workflow Net [8], the Generalized Stochastic Workflow Net is defined.

Definition 3 (GSWN). The Generalized Stochastic Workflow Net is one special class of the Generalized Stochastic Petri Nets satisfying:

a) There are two special places in the GSWN model. One is the unique initiative place i, which has no inputting transitions. The other is the unique end place o, which has no outputting transitions.

b) In order to ensure that the state marking process is a continuous stochastic process with no absorbing states which has the stability solution, a special transition t^* is inserted between the places o and i on the assumption that t^* is a exponential transition with the speed λ. As a result, the new process model becomes a strong-connected Petri net.

c) In the initial marking M_0, only the place i has one token, while the other places are empty.

The condition c) implies that the overall performance of the workflow model in the form of GSWN can be deduced by analysing only a single workflow instance. An assumption that the resources are infinite or available at any time is hidden in this condition. This is somewhat reasonable in reality because the different workflow instances will influence each other only when the resources are conflicting. However, the main resource in the business management process is human. Unlike the manufacturing resources in the manufacturing process which are exclusive at the same time, human is quite flexible and can deal with several activities simultaneously. By neglecting the influence of resource allocation and load, we can remarkably simplify the computation process and rapidly obtain the performance index with an important referenced value.

The marking process of the Generalized Stochastic Petri Net is a Semi-Markov Chain with a discrete state space [10]. The states in the Embedded Markov Chain (EMC) model of the Semi-Markov Chain are corresponded with the markings in the GSPN model. In the state graph, the states are connected by the directed arcs. The mean firing time λ of the transition that results in the state transformation, and the probability p of the random switch are labelled as $\lambda(p)$ close to the arcs. Take 10(0.5) as an example, it means the firing speed is 10, that is, the state will transform for 10 times in unit time; the firing probability of the transition is 0.5. For all the conflicting transitions behind the "OR" structure, their firing probability are smaller than 1.

For the marking process of a GSWN model, the residence time of the vanishing markings is zero, thus only the tangible markings need to be considered. Supposing that there are s tangible markings in the GSWN, $M_1, M_2,\ldots,M_s$; the stability probability of M_i in the marking process is π_i such that $\sum_{i=1}^{s}\pi_i = 1$; the mean residence time of M_i is denoted as m_i, then:

$$m_i = \frac{1}{\sum_{t_k \in T_i} F(M_i, t_k)} \tag{1}$$

where T_i is denoted as the set of all enabled transitions in the marking M_i.

Suppose $Y=[y_1,y_2,\ldots y_s]$, where y_i is the ratio that the marking process passes the marking M_i, which can be calculated out by the limiting probability calculation formula of the discrete time stochastic process: [10]

$$\begin{cases} YA = Y \\ \sum_{i=1}^{s} y_i = 1 \\ y_i \geq 0 \quad (i = 1,2,\ldots,s) \end{cases} \tag{2}$$

where A is denoted as the transition probability matrix.

Therefore, the stability probability π_i can be computed through the following formula:

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$$\pi_i = \frac{y_i m_i}{\sum_{j=1}^{s} y_j m_j} \qquad (i = 1, 2, ..., s) \tag{3}$$

III. Calculation of the Mean Flow time

One of the important time performance measures of business process model is its mean flow time $\overline{T}$, that is, the mean throughout time of all workflow instances. It can be calculated according to the Little rule [10], $\overline{T} = \overline{N} / \nu$, where $\overline{N}$ is the average token number of the Petri net system in its stabile state. The so-called stabile state is denoted as the time when the number of tokens entering into the system is equal to that leaving from the system. ν is the mean number of tokens entering into the net system in every unit time.

Theorem 1. The mean flow time of the GSWN is calculated by $\overline{T} = \frac{1-\pi_o}{\lambda \pi_o} = \frac{\sum_{i=1}^{n} y_i m_i}{y_o}$, where π_o stands for the stability probability at the end marking M_o, λ is the firing speed of the transition t^*, m_i (i=1,2,…,n) is the mean sojourning time of all the tangible markings M_i except the end marking M_o, y_i (i=1,2,…,n) is the ratio that all tangible marking M_i except the end marking M_o being accessed during the marking process, and y_o is the ratio that the end marking M_o being accessed during the marking process.

Proof. Given a child system in the GSWN, $PN'=(P',T',F')$ where $P'=P-\{o\}$, $T'=T-\{t^*\}$, and $F'=F-\{(o,t^*),(t^*,i)\}$, it is obvious that the mean time of tokens passing the child system PN' is just the mean flow time of workflow instances. In the stability state, the number of tokens entering into the child system is equal to that leaving from place o in unit time. Thus $\nu = \lambda \pi_o$. In addition, since only a single workflow instance is taken into account, in the stability state, the mean number of tokens in child system PN' is equal to the probability value that the token is located in PN', thus $\overline{N} = 1 - \pi_o$. According to the Little rule,

$$\overline{T} = \frac{\overline{N}}{\nu} = \frac{1-\pi_o}{\lambda \pi_o} \tag{4}$$

By use of the stability probability calculation formula (3) of GSWN, we draw a conclusion that

$$\pi_o = \frac{y_o m_o}{\sum_{i=1}^{n} y_i m_i + y_o m_o} = \frac{y_o / \lambda}{\sum_{i=1}^{n} y_i m_i + y_o / \lambda} \tag{5}$$

Using (4) and (5), Theorem 1 is proven as follows.

$$\overline{T} = \frac{1-\pi_o}{\lambda \pi_o} = \frac{1 - \dfrac{y_o / \lambda}{\sum_{i=1}^{n} y_i m_i + y_o / \lambda}}{\dfrac{y_o}{\sum_{i=1}^{n} y_i m_i + y_o / \lambda}} = \frac{\sum_{i=1}^{n} y_i m_i}{y_o} \tag{6}$$

Inference 1. The mean flow time $\overline{T}$ has nothing to do with the firing speed λ of the transition t^*.

Proof. From Theorem 1 we can see that the parameter λ is not included in the formula of $\overline{T}$ and all other parameters have nothing to do with λ. So this inference is correct.

IV. Case Study

In this section we will employ the proposed approach to analyze a practical business process in a manufacturing company and obtain its time performance.

A. Directed Graph Model of the Manufacturing Document Verification Process

Fig. 1 shows the manufacturing document verification process of a telecommunication equipment manufacturer in the form of the directed graph. The main purpose of this workflow is to verify the procedure regulation of all kinds of product lines and products, and to submit the document to the sector leaders for signing and releasing. It consists of nine elemental tasks, $A1,\ldots,A9$. The detailed process is explained as follows.

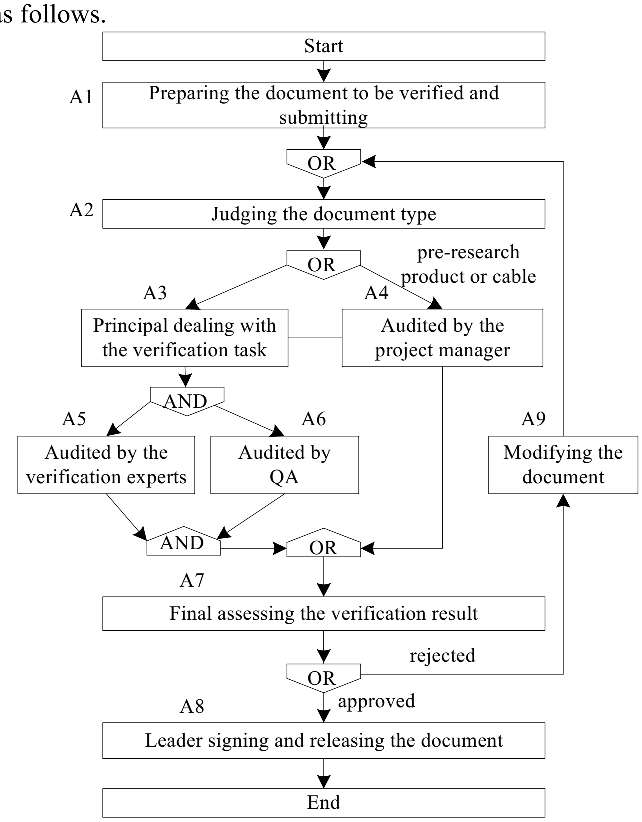

Fig. 1 Directed graph of the manufacturing document verification process.

a) After a workflow instance is started, at the time of task $A1$, the product engineering prepares the original document, fills the application sheet for verification, and then submits this application sheet in the workflow system.

b) At the time of task $A2$, the document type is automatically judged by the workflow system. If the document belongs to the pre-research product or the cable, task $A4$ is enabled and the document is directly submitted to

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the project manager for auditing in step d). Otherwise, a verification team should be organized to deal with this activity in step c).

c) The principal of the verification team receives the task (*A*3) and starts two parallel verification activities, one is done by the verification experts (*A*5) and the other is done by QA (*A*6). These verification results are then summarized.

d) The above verification results from the project manager or the verification team are joined and submitted to final auditing (*A*7). If the document passes the verification, the team leader approves and releases it to the manufacturing shop floor (*A*8). The workflow instance is finished. Otherwise, the application is rejected to the applicant to modify the document (*A*9), and the re-submission is necessary.

B. *Generalized Workflow Net Model*

The left part of Fig. 2 is the GSWN model of the workflow shown in Fig. 1, in which the transitions filled in black stand for the immediate transitions as $T_I = \{t2, t5, t8, t^*\}$, while the transitions with no fillings are the exponential transitions as $T_E = \{t1, t3, t4, t6, t7, t9, t10, t11\}$.

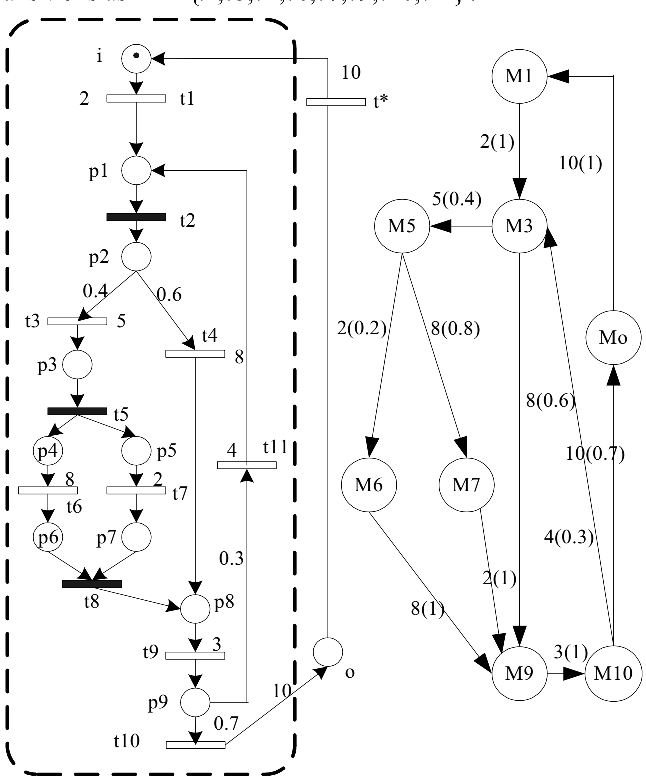

Fig. 2 GSWN model and the state graph of Semi-Markov Chain.

It is assumed that the fire speeds of all transitions are λ_1=2, λ_2=∞, λ_3=5, λ_4=8, λ_5=∞, λ_6=8, λ_7=2, λ_8=∞, λ_9=3, λ_{10}=10, λ_{11}=4, $\lambda t^*=\lambda$, respectively. (The unit is piece per day. The value ∞ represents that the firing time of the immediate transition is zero) According to Inference 1, the value of λ has nothing to do with the calculation result of time performance, so we let λ be 10 for the simple of computing. The randomly switch S is a set such that $S = \{<t3, t4>, <t10, t11>\}$, where $S_{t3} = 0.4$, $S_{t4} = 0.6$, $S_{t10} = 0.7$ and $S_{t11} = 0.3$. That is, the probability that the document belongs to the pre-research product or the cable is 0.6, and the one-pass probability of the applied document for verification is 0.7.

The above GSWN model is identical to a Semi-Markov Chain, of which all the reachable state markings are listed in TABLE I. There are 11 markings, where *M*1, *M*3, *M*5, *M*6, *M*7, *M*9 and *M*o are tangible ones and the others are vanishing ones. The corresponding Semi-Markov Chain is shown in the right part of Fig. 2. Since the sojourning time of the vanishing markings are zero, we delete them from Fig. 2, which may not influence the analysis result at all.

TABLE I
All Reachable State Markings of the GSWN Model in Fig. 2

	Marking Type	i	P1	P2	P3	P4	P5	P6	P7	P8	P9	o
*M*1	Tangible	1	0	0	0	0	0	0	0	0	0	0
*M*2	Vanishing	0	1	0	0	0	0	0	0	0	0	0
*M*3	Tangible	0	0	1	0	0	0	0	0	0	0	0
*M*4	Vanishing	0	0	0	1	0	0	0	0	0	0	0
*M*5	Tangible	0	0	0	0	1	1	0	0	0	0	0
*M*6	Tangible	0	0	0	0	1	0	0	1	0	0	0
*M*7	Tangible	0	0	0	0	0	1	1	0	0	0	0
*M*8	Vanishing	0	0	0	0	0	0	1	1	0	0	0
*M*9	Tangible	0	0	0	0	0	0	0	0	1	0	0
*M*10	Tangible	0	0	0	0	0	0	0	0	0	1	0
*M*o	Tangible	0	0	0	0	0	0	0	0	0	0	1

C. *Calculation of the Stability Probability of the Semi-Markov Chain*

The transition probability matrix A of the Embedded Markov Chain of the Semi-Markov Chain is shown as follows.

$$A = \begin{bmatrix} M_{11} & M_{13} & M_{15} & M_{16} & M_{17} & M_{19} & M_{110} & M_{1o} \\ M_{31} & M_{33} & M_{35} & M_{36} & M_{37} & M_{39} & M_{310} & M_{3o} \\ M_{51} & M_{53} & M_{55} & M_{56} & M_{57} & M_{59} & M_{510} & M_{5o} \\ M_{61} & M_{63} & M_{65} & M_{66} & M_{67} & M_{69} & M_{610} & M_{6o} \\ M_{71} & M_{73} & M_{75} & M_{76} & M_{77} & M_{79} & M_{710} & M_{7o} \\ M_{91} & M_{93} & M_{95} & M_{96} & M_{97} & M_{99} & M_{910} & M_{9o} \\ M_{101} & M_{103} & M_{105} & M_{106} & M_{107} & M_{109} & M_{1010} & M_{1o} \\ M_{o1} & M_{o3} & M_{o5} & M_{o6} & M_{o7} & M_{o9} & M_{o10} & M_{oo} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.4 & 0 & 0 & 0.6 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0.7 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Assuming the limiting probability of EMC is $Y = [y_1\, y_3\, y_5\, y_6\, y_7\, y_9\, y_{10}\, y_o]$, according to (2), we get

$$\begin{cases} YA = Y \\ \sum y_i = 1 \\ y_i \geq 0 \end{cases} \Leftrightarrow \begin{cases} Y(A - E) = 0 & (7) \\ Y[1 \ldots 1]^T = 1 & (8) \\ y_i \geq 0 \end{cases}$$

where $i = 1, 3, 5, 6, 7, 9, 10, o$.

Using (7) and (8), the following equation is deduced.

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$$Y\begin{bmatrix} 1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0.4 & 0 & 0 & 0.6 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0.2 & 0.8 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 1 & 0 & 0.3 & 0 & 0 & 0 & 0 & -1 & 0.7 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (9)$$

Solving (9) in Matlab, the solution is

$$Y = [0.1346\ 0.1923\ 0.0769\ 0.0154\ 0.0615\ 0.1923\ 0.1923\ 0.1346]$$

Let m_i be the mean sojourning time of the marking process M_i satisfying $m_i \in \{m_1\, m_3\, m_5\, m_6\, m_7\, m_9\, m_{10}\, m_o\}$, from (1) we get: $m_1=1/\lambda_1$, $m_3=1/(\lambda_3{*}0.4+\lambda_4{*}0.6)$, $m_5=1/(\lambda_6+\lambda_7)$, $m_6=1/\lambda_6$, $m_7=1/\lambda_7$, $m_9=1/\lambda_9$, $m_{10}=1/(\lambda_{10}{*}0.7+\lambda_{11}{*}0.3)$, $m_o=1/\lambda$, thus the set $\{m_i\}$ is $\{1/2\ \ 1/6.8\ \ 1/10\ \ 1/8\ \ 1/2\ \ 1/3\ \ 1/8.2\ \ 1/10\}$.

Finally, according to (3) we solve the stability probability of Semi-Markov Chain as: $\pi_1 = 0.2840$, $\pi_3 = 0.1194$, $\pi_5 = 0.0324$, $\pi_6 = 0.0081$, $\pi_7 = 0.1297$, $\pi_9 = 0.2704$, $\pi_{10} = 0.0990$, $\pi_o = 0.0566$.

D. *The Mean Flow Time of Workflow*

For the GSWN model in the left part of Fig. 2, the region surrounded by the dashed line is the child system *PN'*, according to Theorem 1, the mean flow time is calculated:

$$\overline{T} = \frac{1-\pi_o}{\lambda\pi_o} = \frac{1-0.0566}{10\times 0.0566} = \frac{0.9434}{0.566} = 1.667\ \text{(day)}.$$

In order to verify if the above result is correct, we set up a simulation model and compare it with the simulation results as shown in Fig. 3.

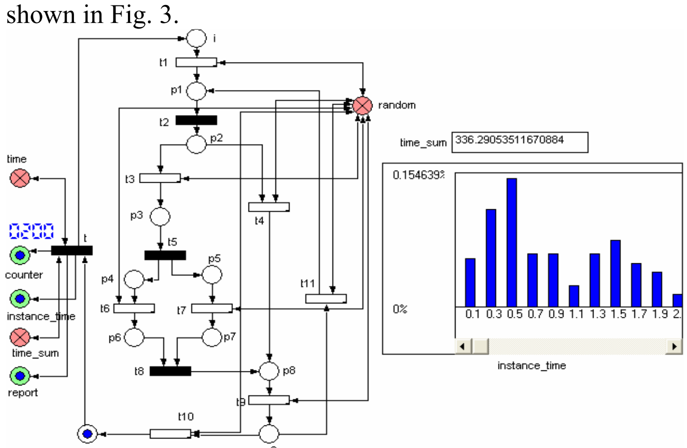

Fig. 3 Process simulation model on ExSpect.

The simulation tool is ExSpect, a powerful simulation tool for modelling, monitoring and analysing based on high-level Petri net. After the running for 200 times, we obtain the total flow time, 336.29. So the mean flow time of a single workflow instance is equal to 336.29/200=1.681(day). Compared with the computed value (1.667 day), the relative error is only 0.8%. Therefore, we conclude that the computed value is very precise.

V. Conclusion

On the assumption that the processing time of all activities in the business process model is either zero or a stochastic variable obeying the exponential distribution, the time performance measure is obtained by use of the GSWN model. The comparative experiment with simulation shows that this approach is effective. However, it should be noted that the Petri net model is confronted with the difficulty of state space explosion when the model becomes complex, so in the future work, the simplification rule of Petri net should be taken into account.

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